

TASHKENT HOUSE PRICE PREDICTOR

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PROJECT IN ONE LINE

Estimate a historical 2019
Tashkent apartment
asking price from
observable listing
attributes.

Reference estimate — not an appraisal

6,725

DE-DUPLICATED ROWS

RF

SELECTED MODEL

0.745

TEST R^2

\$10,573

TEST MAE

WHO NEEDS IT — AND WHAT DOES ML DO?

Lock the user, input, output, and boundary in under 45 seconds.

● USER

Buyers, sellers, agents, and analysts who need a consistent historical reference estimate.

● SUPERVISED REGRESSION

Inputs: district, size, rooms, apartment level, building levels, latitude, longitude.

Target: listing price.

● OUTPUT + BOUNDARY

One estimated 2019 asking price in USD, plus warnings.

Not a current quote, sale guarantee, or legal appraisal.

RAW APARTMENT → VALIDATED PIPELINE → USD ESTIMATE +
WARNING

FROM 7,421 LISTINGS TO A LEAKAGE-SAFE TEST

CC0 uybor.uz advertisements collected in Tashkent in 2019.



Leakage controls

duplicate removal before split • no target-derived features • encoding/scaling inside each sklearn pipeline • test not used for selection

FOUR APPROACHES — ONE EVIDENCE-BASED CHOICE

Final model selected by the lowest five-fold development CV MAE, not test performance.

MODEL	CV MAE	CV RMSE	CV R ²
Median baseline	\$25,220	\$47,276	-0.081
Log Ridge	\$15,580	\$47,449	-0.248
Random Forest	\$11,108	\$23,289	0.736
Gradient Boosting	\$12,099	\$25,473	0.685



WHY RANDOM FOREST?

Best validation MAE

Captures non-linear size
× location interactions

Trade-off: larger and less
interpretable than Ridge

UNSEEN-TEST RESULT — WITH THE FAILURE VISIBLE

The baseline comparison and error case matter more than a polished success claim.

\$10,573

MAE

\$23,162

RMSE

0.745

R^2

16.66%

MAPE

−56.3%

MAE vs baseline



LARGEST ERROR

Mirobod • 456 m² • 10 rooms

Actual: \$800,000 Predicted: \$354,962

Absolute error: \$445,038



WHAT IT TEACHES

Luxury premiums and condition are not fully observed. Bektemir and Yangihayot are too sparse for reliable slice claims.

ONE REAL INPUT → ONE VISIBLE RESULT

Colab route: setup → load saved pipeline → predict → show validation error.

RAW INPUT

District Chilonzor
Size 70 m²
Rooms 3
Level 3 / 5
Coordinates 41.3002, 69.2108



ESTIMATED 2019 ASKING PRICE

\$53,532 USD

EDGE CASE

Level 9 / max 5 → clear ValueError

Open: [README Colab badge](#) • [Run all](#) • Do not tour source code unless asked

HONEST LIMITS — CLEAR NEXT STEP

A useful educational model is stronger when its boundary is explicit.

LIMITATIONS

2019 asking prices
No condition / building age /
legal status
Sparse districts
Location bias and market
drift

SAFE USE

Historical reference only
Compare with recent listings
Require human review
No lending, tax, or legal
appraisal

NEXT IMPROVEMENT

Collect current transaction
data with date, condition,
building year, renovation,
and legal status; then use a
time-aware holdout.

Thank you. I am ready for one evidence-based question.

OFFICIAL CRITERIA → EXACT REPOSITORY PROOF

Open docs/evidence_matrix.md for the complete claim → location → why-proof mapping.

1 Problem

README + Project Brief

2 Data

data/README + EDA +
src/data.py

3 Models

model_comparison.csv +
src/model.py

4 Evaluation

test metrics + errors +
district slices

5 Delivery

demo.ipynb + saved joblib
+ predict.py

6 Reproduce

README +
clean_run_check + CI

7 Responsible

README limitations +
data card

8 Defense

deck + pitch + question
bank

YELLOW only where external approval/rehearsal evidence is still required.

THE SAME PIPELINE TRAINS AND PREDICTS

Preprocessing is serialized with the estimator, preventing notebook-only inference drift.

● RAW INPUT

district • rooms • size • level •
max_levels • lat • lng

● VALIDATE

required fields • positive values •
 $\text{level} \leq \text{max_levels}$

● FEATURE

$\text{floor_ratio} = \text{level} / \text{max_levels}$

● PREPROCESS

OneHotEncoder district • numeric
passthrough / scaling

● MODEL

selected Random Forest inside
sklearn Pipeline

● OUTPUT

USD estimate + unseen/range
warnings

Source proof: [src/data.py](#) + [src/model.py](#) + [artifacts/house_price_pipeline.joblib](#)

ANSWER WITHOUT GUESSING

Direct answer → exact evidence → limitation or next step.

● WHY RANDOM FOREST?

Lowest CV MAE (\$11,108).
Open
reports/model_comparison.csv.
Trade-off: size and
interpretability.

● IS IT CURRENT?

No — 2019 asking-price
estimate.
Open README limitations.
Next: current transaction data
+ time split.

● BIGGEST FAILURE?

\$800k Mirobod actual vs
~\$355k predicted.
Open largest_errors.csv.
Missing luxury/condition
signal.

Full evidence-anchored answers: docs/defense_question_bank.md